Chemistry

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp} = 0°C
T_{bp} = 100°C
c_{ice} = 2.09 J/g·K
c_{water} = 4.184 J/g·K
c_{steam} = 2.03 J/g·K
ΔH_{fus} = 334 J/g
ΔH_{vap} = 2260 J/g
K_f = 1.86 °C/*m*
K_b = 0.512 °C/*m*

Constants

R = 0.08206 L·atm/mol·K
R = 8.314 J/mol·K
R = 62.36 L·torr/mol·K
$e = 1.602 \times 10^{-19}$ C
$N_A = 6.022 \times 10^{23}$ mol^{-1}
$k = 1.38 \times 10^{-23}$ J/K
$h = 6.626 \times 10^{-34}$ J·s
$c = 3.00 \times 10^{8}$ m/s
$R_H = 2.178 \times 10^{-18}$ J
$m_e = 9.11 \times 10^{-31}$ kg

Activity Series of Metals in Aqueous Solution

Metal	Oxidation Reaction
Lithium	$Li(s) \rightarrow Li^+(aq) + e^-$
Potassium	$K(s) \rightarrow K^+(aq) + e^-$
Barium	$Ba(s) \rightarrow Ba^{2+}(aq) + 2e^-$
Calcium	$Ca(s) \rightarrow Ca^{2+}(aq) + 2e^-$
Sodium	$Na(s) \rightarrow Na^+(aq) + e^-$
Magnesium	$Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$
Aluminum	$Al(s) \rightarrow Al^{3+}(aq) + 3e^-$
Manganese	$Mn(s) \rightarrow Mn^{2+}(aq) + 2e^-$
Zinc	$Zn(s) \rightarrow Zn^{2+}(aq) + e^-$
Chromium	$Cr(s) \rightarrow Cr^{3+}(aq) + 3e^-$
Iron	$Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$
Cobalt	$Co(s) \rightarrow Co^{2+}(aq) + 2e^-$
Nickel	$Ni(s) \rightarrow Ni^{2+}(aq) + 2e^-$
Tin	$Sn(s) \rightarrow Sn^{2+}(aq) + 2e^-$
Lead	$Pb(s) \rightarrow Pb^{2+}(aq) + 2e^-$
Hydrogen	$H_2(g) \rightarrow 2\ H^+(aq) + 2e^-$
Copper	$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$
Silver	$Ag(s) \rightarrow Ag^+(aq) + e^-$
Mercury	$Hg(l) \rightarrow Hg^{2+}(aq) + 2e^-$
Platinum	$Pt(s) \rightarrow Pt^{2+}(aq) + 2e^-$
Gold	$Au(s) \rightarrow Au^{3+}(aq) + 3e^-$